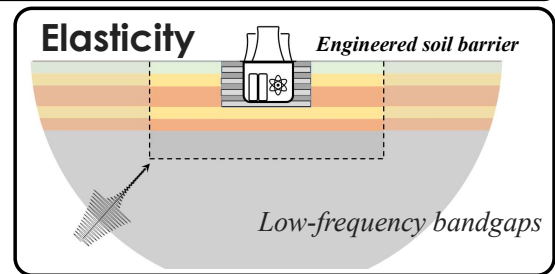
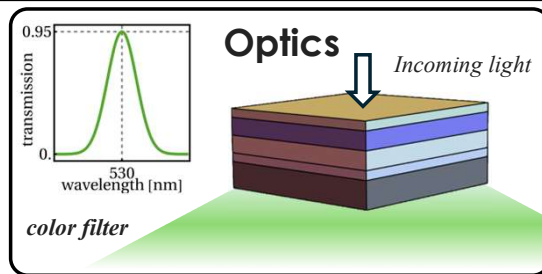
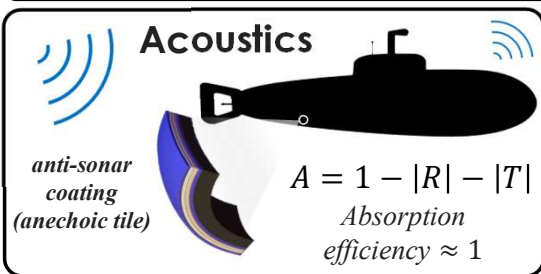


General Structure

$$\frac{d}{dx} \begin{bmatrix} f \\ f' \end{bmatrix} = \begin{bmatrix} 0 & \alpha \\ \beta & 0 \end{bmatrix} \begin{bmatrix} f \\ f' \end{bmatrix} \rightarrow \mathbf{f}(x_{k+1}) = \underbrace{\exp(\mathbf{A}_k l)}_{\mathbf{T}_k(x_{k+1}, x_k) \in \mathbf{SL}_2(\mathbb{R})} \mathbf{f}(x_k) \rightarrow \mathbf{f}(x_N) = \overbrace{\mathbf{T}_N \dots \mathbf{T}_1}^{\text{multiple layers}} \mathbf{f}(x_0)$$

one homogeneous layer



“Transfer matrices possess harmonic structure”

Garcia-Suarez (2022) JMPS

$$\mathbf{T}_{11} = \sum_k \mathbf{J}_k \cos(\boldsymbol{\tau}_k \omega)$$

Amplitudes ($\mathbf{J}_k$) and periods($\boldsymbol{\tau}_k$) known in closed form as a function of layer **thicknesses** and layer **material properties**